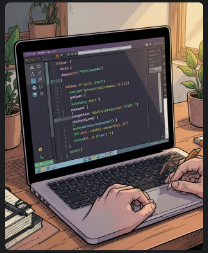
Data Analytics Internship Portfolio

Presented by Naguru Girija Rani — Data Analytics Intern

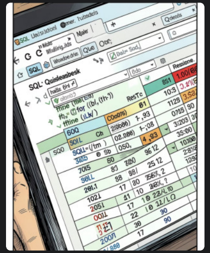
Date: August 2026

SKILLS



Python

Data manipulation, scripting, and automation with Pandas & NumPy.



SQL

Complex joins, aggregations, and KPI calculations for business reporting.

Power BI

Interactive dashboards with filters, KPI cards, and drill-throughs.

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About Me

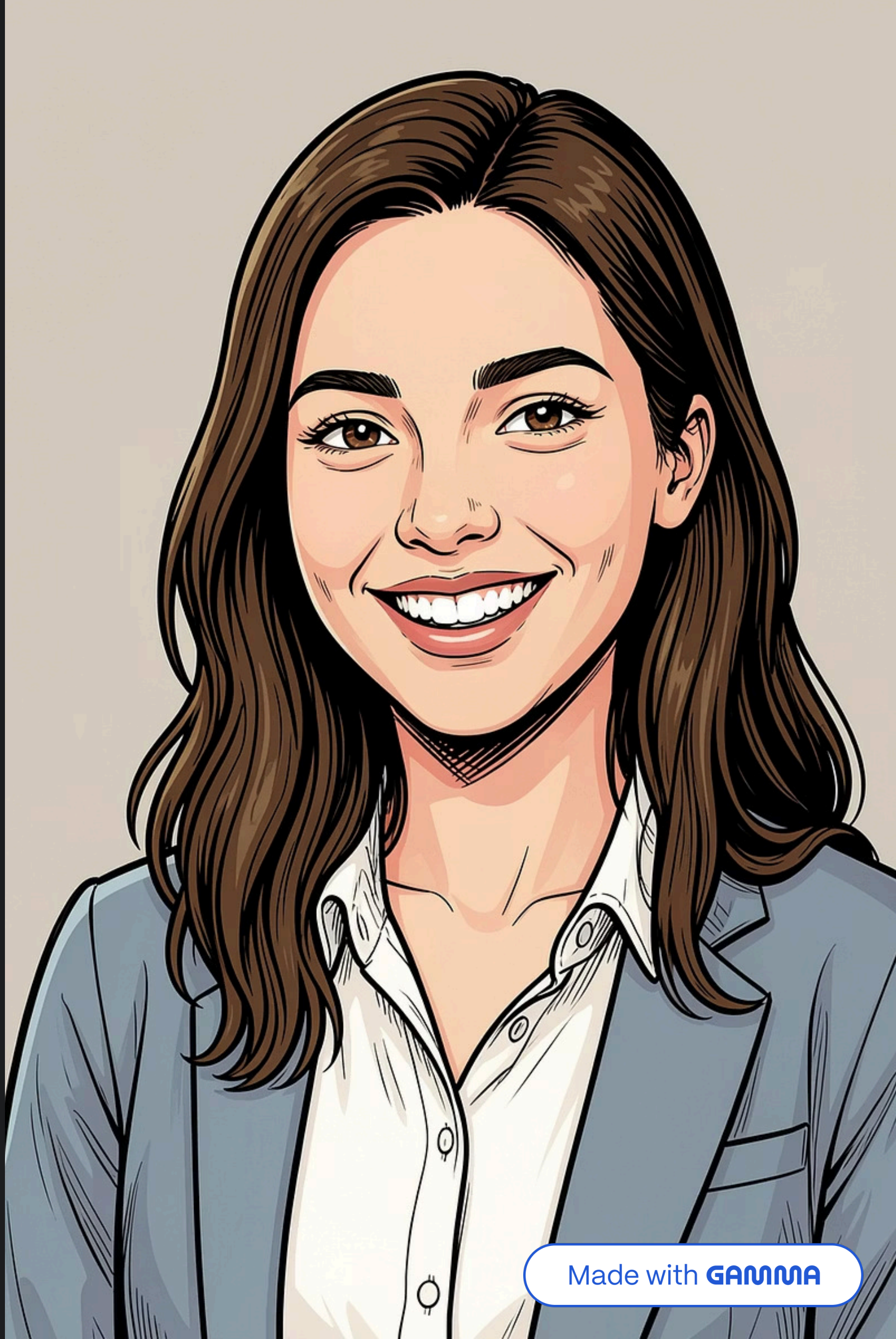
Hello — I'm Girija Naguru, an aspiring data analyst focused on turning messy data into clear, actionable insights. I approach problems with curiosity, careful validation, and a bias toward measurable impact.

Core Strengths

Data cleaning, exploratory analysis, SQL, dashboarding, and storytelling.

Career Objective

To grow as a data professional by applying analytical rigor and technical skills to real business problems.



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Internship Overview

The internship was structured to expose me to the full analytics lifecycle: data ingestion, cleaning, analysis, modeling, and dashboard delivery. Work prioritized business-relevant questions and reproducible code.

Objectives

- Understand analytics lifecycle
- Work with real datasets
- Deliver insights and dashboards

Tools & Libraries

- Python, Pandas, NumPy
- Matplotlib, Seaborn
- SQL, Power BI, GitHub

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Project 1 — Data Cleaning

Objective: Prepare raw datasets for reliable analysis by improving quality and consistency.

Key Activities

- Removed duplicates and fixed incorrect types
- Imputed or removed missing values using context-driven rules
- Standardized date and categorical formats
- Documented transformations for reproducibility

Outcome

Delivered a clean, validated dataset and automated cleaning scripts that reduced downstream errors and saved analyst time.

Visual: Data cleaning screenshot attached for review.

A	B	C	D	E	F	G	H
customer_id	name	dob	city	purchase_date	amount	age	spending_category
101	Anita	21-05-1998	hyderabad	10-01-2024	250	28	Low
102	Ravi	15-08-2000	vizag	10-02-2024	500	26	Low
103	Sneha	19-07-1992	chennai	05-03-2024	700	34	High
104	Kiran	10-12-1995	hyderabad	10-01-2024	250	31	Low
105	Amit	19-07-1992	unknown	12-04-2024	1200	34	High

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Project 2 — Exploratory Data Analysis (EDA)

Objective: Discover trends, relationships, and risks to inform business decisions.

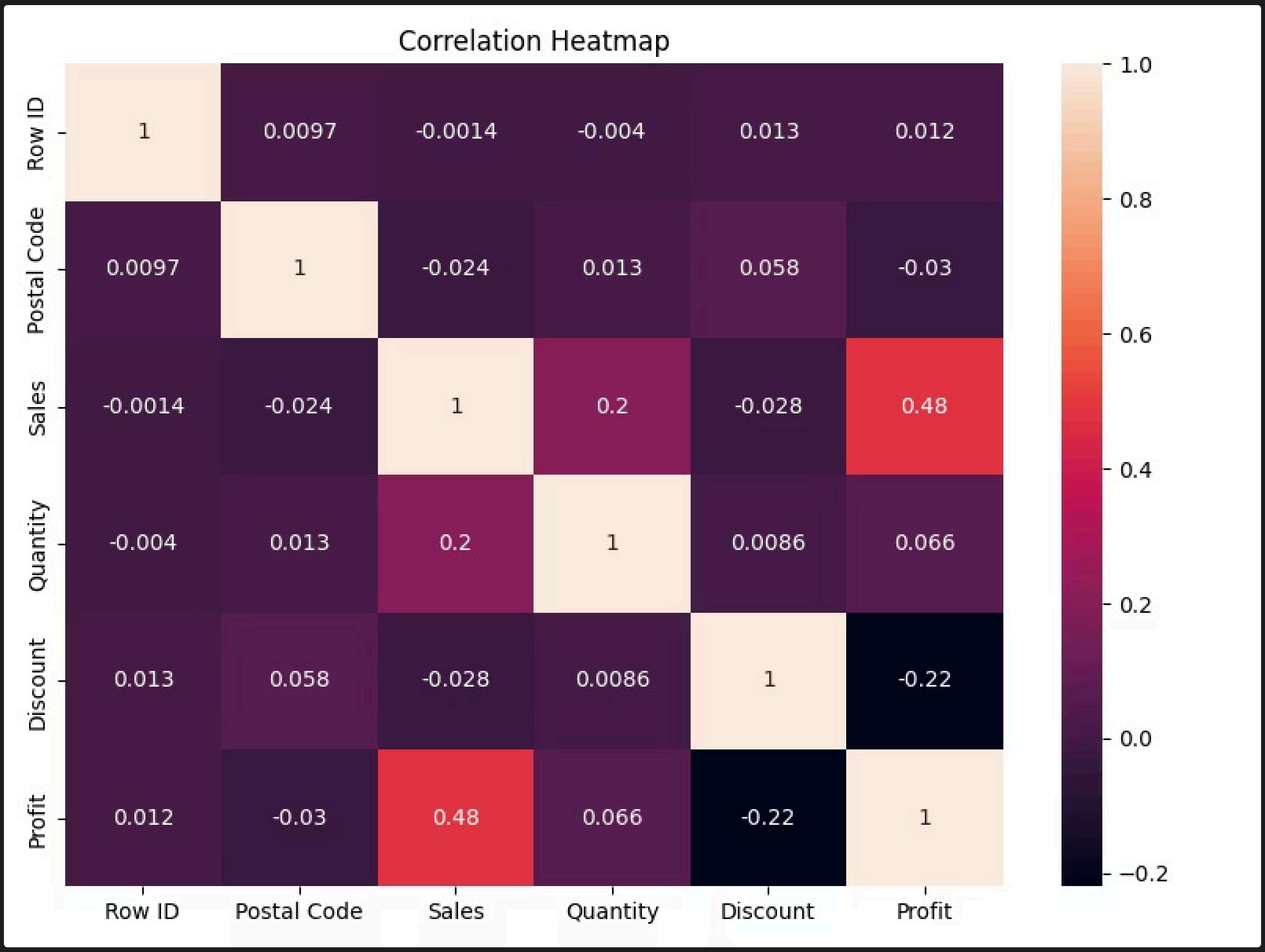
Approach

- Univariate and bivariate distributions
- Correlation matrices to identify predictive features
- Outlier detection and contextual investigation
- Visual storytelling to surface actionable patterns

Visualizations

- Histograms & density plots for distributions
- Bar charts for categorical comparison
- Heatmaps for correlation insights
- Time-series trend charts for seasonality

Outcome: Identified high-impact features and recommended next-step analyses for stakeholders.



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Project 3 — SQL Analysis

Objective: Extract, transform, and analyze data to calculate business KPIs and prepare reporting tables.

01

Common Tasks

Filtering, joins across source tables, GROUP BY aggregations, window functions for running metrics.

02

Key Metrics

Total Revenue, Orders, Customer Growth, Average Order Value, Product Performance.

03

Outcome

Created reproducible SQL scripts and summary tables used directly by Power BI dashboards.

Visual: Example SQL screenshot included in appendix.

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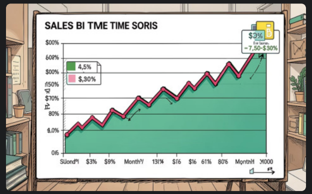
Project 4 — Power BI Dashboard

Objective: Build an interactive dashboard to monitor business performance and enable stakeholder exploration.

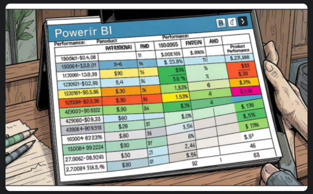
KPI Cards

At-a-glance metrics for leadership: revenue, orders, and customer growth.



Sales Trends

Interactive filters and drill-throughs to investigate seasonality and anomalies.



Product Analysis

Rankings and conditional highlights for product-level decisions.

Outcome: Delivered a polished, user-friendly dashboard for both analysts and executives.

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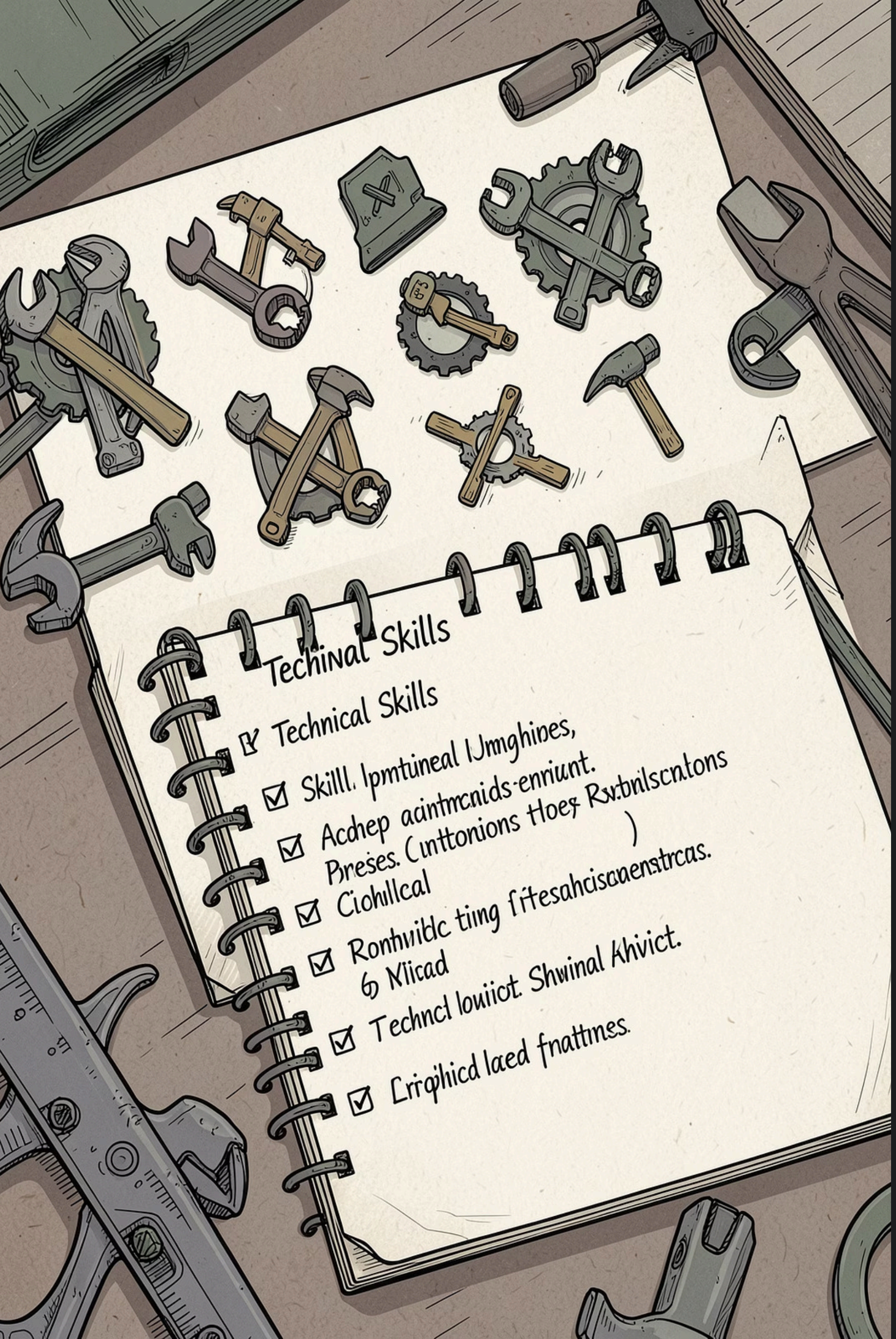

Skills Gained

Technical & Analysis

- Python (Pandas, NumPy)
- SQL — complex queries & window functions
- EDA with Matplotlib & Seaborn

Visualization & Tools

- Power BI — interactive dashboards
- Version control with Git & GitHub
- Reproducible workflows and documentation

These skills improved both technical delivery and the ability to communicate findings clearly to stakeholders.

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Challenges & Solutions

1

Challenge 1: Missing & inconsistent data

Solution: Applied context-aware imputation, type correction, and validation rules to restore data integrity.

2

Challenge 2: Complex dataset structure

Solution: Performed layered EDA, built summary tables, and documented entity relationships for clarity.

3

Challenge 3: Communicatin g insights

Solution: Designed targeted Power BI views and concise storytelling slides for executive decision-making.

Result: Clean, actionable deliverables used in regular stakeholder reviews.

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Conclusion & Thank You

This internship delivered practical end-to-end experience across data cleaning, EDA, SQL reporting, and dashboard development. I built reproducible workflows and delivered business-ready insights.

Key Achievements

- Transformed raw data into validated datasets
- Produced EDA that informed business priorities
- Built an interactive Power BI dashboard for monitoring

Next Steps

Continue refining dashboard features, implement automated data pipelines, and apply predictive models where appropriate.

Contact

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GitHub: https://github.com/dashboard

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